

DERIVATIVES: FORWARD, FUTURES AND SWAPS

(DEFINITIONS, TYPES, PRICING, TRADING)

Source:

- *Options, Futures and Other Derivatives*, Hull; Ch. 2, 3, 5, 7
- *Investments*, BKM; Part VI; Chapter 23.4,

Derivatives are securities whose prices are determined by, or derived from, the prices of other securities. Thus, they are powerful tools for both hedging and speculation.

Definitions and types

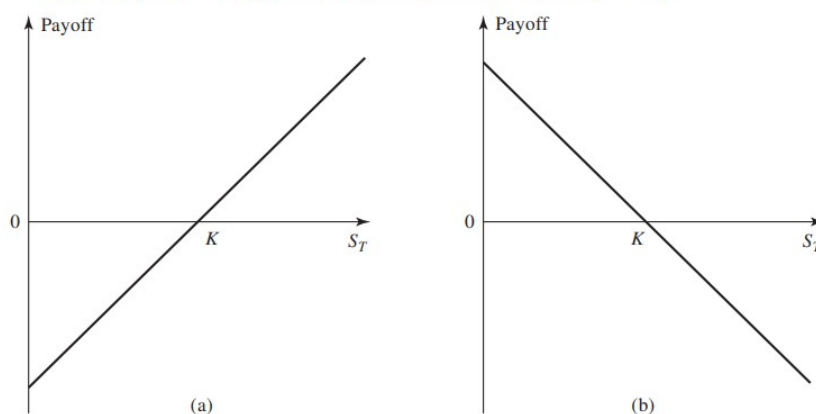
Forward contract: An agreement to buy or sell an asset at a certain future time for a certain price. Forward contracts are traded on the over-the-counter market. This allows flexibility but also exposes each party to counterparty risk. Commonly used by businesses that need to hedge against foreign exchange or commodity price risk in specific quantities and timeframes.

The party who has the *long position* agrees to buy the underlying asset on a specified future date for a specified price. The one in *short positions* agrees to sell it.

Payoff of a long forward contract: $S(T) - K$, where K is the delivery price, $S(T)$ is the spot price of the asset at maturity of the contract. The person in the long position is **obligated** to buy the asset worth $S(T)$ for K . Payoff for the short position is $K - S(T)$.

It costs nothing to enter the contract, so payoff is the traders total gain or loss.

Figure 1.2 Payoffs from forward contracts: (a) long position, (b) short position.
Delivery price = K ; price of asset at contract maturity = S_T .



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Short selling: Short selling refers to the trading technique when an investor is borrowing and selling shares (selling shares, which are not owned, at the current price) and at some later stage, she will close out the position by purchasing the same number of shares for the respective price on the spot market. An investor with a short position must pay the broker (which will transfer them to the owner) any income, such as dividends or interest. That is a 'cost' of the short trader (additionally to the fee for borrowing). The investors are required to maintain a *margin account* (consisting of cash and marketable securities) to guarantee the investor will not walk away from the short position if things go wrong. There is an *initial margin*, and an *additional margin* may be required, if prices keep increasing, otherwise the position is closed.

There are few assumptions regarding *some market participants* (like large derivative dealers) that must be true for the theory to work:

- *No transaction costs* when they trade
- The *same tax rate* on all net trading profits
- Borrow and lend money at the same r_f rate
- *Take advantage of arbitrage opportunities* as they occur

Notations:

$F(0)$ = forward/futures price that would be applicable if we negotiated the contract today

$S(0)$ = underlying asset price today

T = maturity / time until delivery date

r = zero-coupon r_f rate

Pricing of Forward Contracts

Price of a forward contract on an asset with no income (the easiest) ex. non-dividend paying stocks or zero-coupon bonds:

$$F_0 = S_0 e^{rT}$$

Arbitrage opportunities with mispricing:

If $F_0 > S_0 e^{rT}$, arbitrageurs can buy the asset and short forward contracts on the asset. If $F_0 < S_0 e^{rT}$, they can short the asset and enter into long forward contracts on it.² In our

Table 5.2 Arbitrage opportunities when forward price is out of line with spot price for asset providing no income. (Asset price = \$40; interest rate = 5%; maturity of forward contract = 3 months.)

Forward Price = \$43	Forward Price = \$39
<i>Action now:</i>	<i>Action now:</i>
Borrow \$40 at 5% for 3 months	Short 1 unit of asset to realize \$40
Buy one unit of asset	Invest \$40 at 5% for 3 months
Enter into forward contract to sell asset in 3 months for \$43	Enter into a forward contract to buy asset in 3 months for \$39
<i>Action in 3 months:</i>	<i>Action in 3 months:</i>
Sell asset for \$43	Buy asset for \$39
Use \$40.50 to repay loan with interest	Close short position
	Receive \$40.50 from investment
Profit realized = \$2.50	Profit realized = \$1.50

A long forward contract and a spot purchase both lead to the *asset being owned at time T*. The forward price of a security providing no income is always higher than the spot price because of the cost of financing the spot purchase of the asset during the life of the forward contract.

Arbitrage if short selling is not possible:

We continue to consider the case where the underlying investment asset gives rise to no storage costs or income. If $F_0 > S_0 e^{rT}$, an investor can adopt the following strategy:

1. Borrow S_0 dollars at an interest rate r for T years.
2. Buy 1 unit of the asset.
3. Enter into a forward contract to sell 1 unit of the asset.

At time T , the asset is sold for F_0 . An amount $S_0 e^{rT}$ is required to repay the loan at this time and the investor makes a profit of $F_0 - S_0 e^{rT}$.

Suppose next that $F_0 < S_0 e^{rT}$. In this case, an investor who owns the asset can:

1. Sell the asset for S_0 .
2. Invest the proceeds at interest rate r for time T .
3. Enter into a forward contract to buy 1 unit of the asset.

At time T , the cash invested has grown to $S_0 e^{rT}$. The asset is repurchased for F_0 and the investor makes a profit of $S_0 e^{rT} - F_0$ relative to the position the investor would have been in if the asset had been kept.

Price of a forward contract with *known income* (perfectly predictable CF):

$$F_0 = (S_0 - I)e^{rT}$$

where I = PV of income during the contract.

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Price of a forward contract with *known yield*:

$$F_0 = S_0 e^{(r-q)T},$$

where q = average yield on the asset during time T .

Valuation of the Forward Contracts

The value of a forward contract at the time it is first entered into is *close to zero* because K is set to be equal to F . Later it can be $\pm$. Banks are valuing contracts each day (*mark to market* activity).

$$f = (F_0 - K)e^{-rT} \qquad f = S_0 - Ke^{-rT}$$

where f = value of forward contract today.

At the beginning of the life of the contract $F(0) = K$, as time passes K remains the same (as by definition is part of the contract), but the forward price changes, and thus the value of the contract.

Derivation of this equation: have a portfolio where we can enter a forward contract to

- 1) buy an underlying asset for K at time T and to
- 2) sell the asset for $F(0)$ at time T .

The two payoffs at T are:

- from 1) $S_T - K$ (at time T , after buying it for K , we can sell it for S_T)
- from 2) $F(0) - S_T$ (at time T we buy it at the spot price S_T and can sell it at the predetermined F_0 price)
- This leads to $F(0) - K$ at time T , and *this payoff is known for sure today*, so we can discount with the risk free rate e^{-rT} .

Formulas for forward contracts (these are just derived from the above ones, by plugging in the formulas for $F(0)$):

$$f = S_0 - Ke^{-rT} \quad f = S_0 - I - Ke^{-rT} \quad f = S_0 e^{-qT} - Ke^{-rT}$$

$$f = S * e^{-rfT} - K * e^{-rT}$$

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A stock index can usually be regarded as the price of an investment asset that pays dividends.

Futures price on stock indices: $F_0 = S_0 e^{(r-q)T}$,

where q is the avg annualized dividend yield.

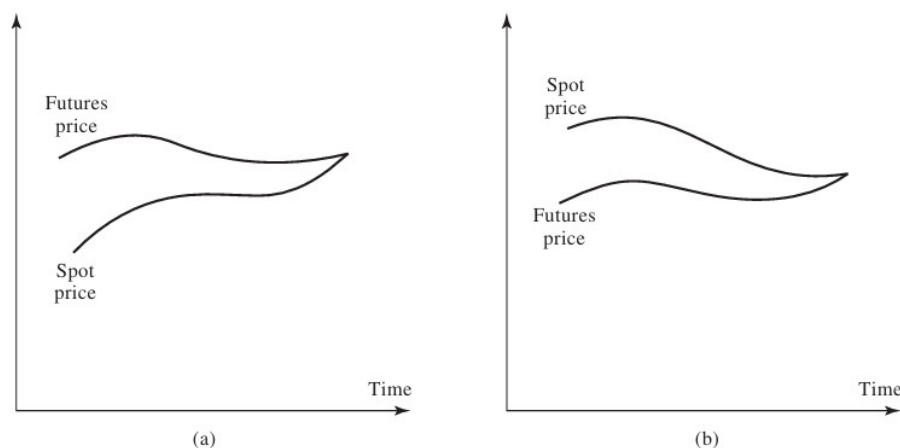
Forward price for currencies: $F_0 = S_0 e^{(r-r_f)T}$,

where r_f is the foreign currency.

Future contracts: The difference compared to forward contracts is that futures are normally *traded on an exchange*. To make trading possible, the exchange specifies certain standardized features of the contract. As the two parties do not necessarily know each other, a *clearinghouse* stands between them. Another huge difference is the *daily settlement / marking to market* feature, meaning that at the end of each trading day the *margin account* is adjusted to reflect the trader's gain or loss.

Most of the (long) futures contracts do not lead to delivery bc traders choose to close out their positions (entering into the opposite trade) prior to the delivery period. The futures price converges to the spot price of the underlying asset as the delivery period is approached.

Figure 2.1 Relationship between futures price and spot price as the delivery period is approached: (a) Futures price above spot price; (b) futures price below spot price.



Futures Trading Process with futures (BKM):

Trades take place through electronic networks. Once a trade is initiated, the *clearinghouse* becomes the seller of the contract for the long position and the buyer of the contract for the short position. The clearinghouse is the trading partner of each trader, and he is the only party that can be hurt by the failure of any trader to satisfy its obligation. The clearinghouse's position is

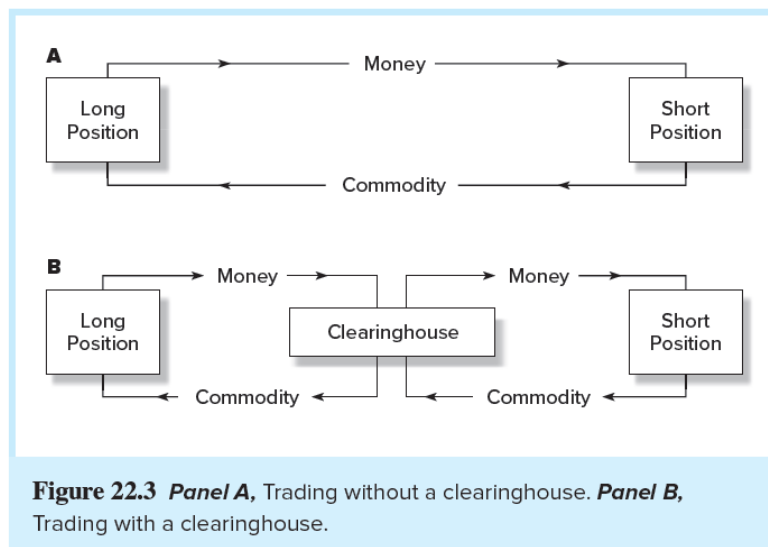
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neutral. It acts as an intermediary for futures transactions, guaranteeing the performance of the parties involved in each transaction. No money changes hands when the transaction is made.

At the time of entering into the transaction, the *initial margin* (5-15% of the contract) must be placed in the margin account (cash and cash-equivalents). All positions recognize profits as they accrue: this daily settling is called *marking to market*. Due to M2M, the value of the margin account changes each day. The daily flow of funds between traders to reflect gains and losses is known as *variation margin*. If the margin account falls below the critical value called *maintenance margin*, the trader receives a *margin call* to refill its balance up to the initial margin. Some contracts call for delivery (commodities) while others for cash settlement (stock index).

The market is regulated by the CFTC (Commodity Futures Trading Commission). Futures exchange may set limits on the amount by which futures prices may change from one day to the next. However, these limits offer no real protection against fluctuations in equilibrium prices. Also, investors do not have control over the tax year in which they realize gains/losses, because they pay them at year-end on cumulated profits/losses.



A *futures contract* specifies the asset, the contract size (amnt of assets delivered under one contract), delivery arrangements (the place of delivery), delivery month, price quotes (dollar, cents etc.), price limit (daily price movements can be limited) and position limit (max # of contracts that can be held).

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Difference between forward and futures contracts:

Table 2.3 Comparison of forward and futures contracts.

<i>Forward</i>	<i>Futures</i>
Private contract between two parties	Traded on an exchange
Not standardized	Standardized contract
Usually one specified delivery date	Range of delivery dates
Settled at end of contract	Settled daily
Delivery or final cash settlement usually takes place	Contract is usually closed out prior to maturity
Some credit risk	Virtually no credit risk

When a futures price changes, the gain or loss on a futures contract is calculated as the change in the futures price multiplied by the size of the position. This gain is realized almost *immediately* because futures contracts are settled daily. When a forward price changes, the gain or loss is the *present value* of the change in the forward price multiplied by the size of the position.

When the short-term r_f rate is constant/known (no uncertainty), forward and futures prices in theory are equal. *When interest rates vary unpredictably, forward and futures prices are no longer the same in theory due to the daily settlement procedure.* (When e.g. S increases, and it is positively correlated with interest rate movements, the futures holder makes immediate gain, and it is likely that interest rates also increased at which gain can be reinvested. Similarly, when S decreases the interest rates are likely to decrease as well, thereby reducing the rate at which the loss must be financed.) Positive correlation works in the investor's favor. *The investor of forward contracts is not affected by interest rate movements.* In this case Futures price > forward price. Other theory assumes negative correlation between S and the interest rate, in which case forward price > futures price. Other differences could be caused by taxes, transaction costs, margin requirements, counterparty risk.

Forward contracts are easier to analyze than futures contracts because there is no daily settlement – only a single payment at maturity. *The forward price and futures price of an asset are usually very close when the maturities of the two contracts are the same.*

Futures prices vs Expected Spot Prices

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There are 3 *traditional* theories that explain how futures prices forecast the expected spot prices.

- Expectations Hypothesis

It states that *futures price equals the expected value of the future spot price*: $F_0 = E(S_T)$. Thus, the expected profit for each side is 0 ($F_0 - E(S_T) = E(S_T) - F_0 = 0$). The hypothesis relies on *risk neutrality*: if all market participants are risk neutral, they should agree on a futures price that provides an expected profit of zero to all parties. Under uncertainty, however, futures price won't equal currently expected spot price, because of *risk premiums*.

- Normal Backwardation

For most commodities there are natural hedgers who want to hedge against market risk by entering short futures (ex. farmers want to fix the selling price of wheat). To induce speculators to enter the opposite side (long position) of the contract, these natural hedgers have to offer them an expectation of profit. Thus, the *futures price has to be below the expected spot price*: $F_0 < E(S_T)$. Hedgers will bear this expected loss to avoid the risk of uncertain commodity prices.

- Contango

For most commodities there are natural hedgers who want to hedge against market risk by entering long futures (ex. grain processors want to fix the purchase price of wheat). To induce speculators to enter the opposite side (short position) of the contract, these natural hedgers have to pay a premium to speculators to enter into the contract. Thus, the *futures price has to be above the expected spot price*: $F_0 > E(S_T)$. Hedgers will bear this expected loss to avoid the risk of uncertain commodity prices.

Hedging Using Forwards/Futures

Hedging refers to techniques that offset particular sources of risk.

Short hedge: If a company needs to *sell a product* at a future date, it can hedge its exposure to price fluctuation by taking a *short futures position*: if the price until T decreases, it sells the product for less on the spot market, but due to the futures contract, it fixed the sell price, so it gains money and can close the contract at price K.

Ex. a company will need to receive GBP in 3-months, so it can enter a 3-month SF to sell that GBP after receipt, thus locking in the other currency.

Long hedge: If a company needs to *purchase a product* at a future date, it can hedge itself by taking a *long futures position*: inversely.

Ex. a company will need to pay GBP in 3-months, so it can enter a 3-month LF to buy GBP, thus locking in the GBP payment.

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Futures contracts can also be used to hedge *currency/exchange risk*. If a company has a payment let's say in 6 months in GBP, it can lock in the exchange rate at the 6-month forward exchange rate now by entering a short forward contract to immediately sell the GBP to USD as it gets them.

Futures may also be used to hedge *market risk*. A fund manager may expect that the market will go up on long run but will be vulnerable to a sharp downturn on the short term. The manager could enter a short future on the stock index: when the portfolio falls in value, the futures contract value will rise in value.

Interest rate futures can be used to hedge *interest rate risk*. A portfolio manager has a bond portfolio with duration x . If interest rates rise, the portfolio value would fall. He could take an offsetting position in an interest rate futures.

Arguments for and against hedging:

- + Corporates with no skills or expertise in predicting variables such as interest rates, exchange rates and commodity prices can lock in in/outflows. So, they avoid unpleasant outcomes.
- Shareholders can do the hedging themselves (ignoring commissions and transaction costs)
- A company that does not hedge can expect its profit margins to be roughly constant, while company with hedging can expect its profit margin to fluctuate.
- Hedging can lead to worse outcomes.

In practice hedging is not always straightforward due to:

- The asset whose price is to be hedged may not be the same as the asset underlying the futures contract.
- There may be uncertainty about the exact date when the asset will be sold or bought.
- The hedge may require the futures contract to be closed out before its delivery month.

These problems are incorporated in the so-called **basis risk**.

$$\text{Basis} = \text{Spot price of asset to be hedged (S)} - \text{Futures price of contract used (F)}$$

Basis should be zero if asset to be hedged and the asset underlying the futures are the same at maturity. Before maturity the basis can fluctuate, called *strengthening of the basis* or *weakening of the basis* (which is again the difference between spot and futures price).

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A company that uses a short hedge because it plans to sell the underlying asset: If the basis strengthens unexpectedly, the company's position improves, as it will get a higher price for the asset after futures gains or losses are considered.

Choose a delivery month that is as close as possible to, but later than, the end of the life of the hedge. When there is no futures contract on the asset being hedged, choose the contract whose futures price is most highly correlated with the asset price. This is **cross hedging**, and it occurs when the asset underlying the futures contract and the asset whose price is to be hedged are different. The **hedge ratio** is the ratio of the size of the position taken in futures contracts to the size of the exposure (if the two assets are the same this ratio is 1).

Minimum Variance Hedge Ratio minimizes the variance of the value of the hedged position.

$$h^* = \rho \frac{\sigma_S}{\sigma_F}$$

and if there is daily settlement:

$$h^* = \hat{\rho} \frac{\hat{\sigma}_S S}{\hat{\sigma}_F F}$$

Optimal Number of Contracts:

Q_A : Size of position being hedged (units)

Q_F : Size of one futures contract (units)

N^* : Optimal number of futures contracts for hedging.

The futures contracts should be on $h^* Q_A$ units of the asset. The number of futures contracts required is therefore given by

$$N^* = \frac{h^* Q_A}{Q_F} \quad (3.2)$$

Forward Rate Agreements

Forward Rate Agreement (FRA): is an agreement to exchange a predetermined fixed rate for a reference rate that will be observed in the market at a future time. Both rates are applied to a specified principal, but the principal itself is not exchanged. The transaction is used to fix the borrowing/lending rate at some future time.

Historically the reference rate was LIBOR, now it is SOFR. FRAs can be valued by *assuming that forward rates are realized*.

If you are long FRA: you *pay the fix* (R_K) and *receive the floating* (R_F).

If you are short FRA: you *pay the floating* (R_F) and *receive the fix* (R_K).

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- $\tau = T(2) - T(1)$
- $R(K)$: fixed rate agreed is what is set in the contract. It is what we can see at $t = 0$ for the period τ . It CAN'T be changed.
- $R(F)$: the current forward rate for the reference rate DF is the discount factor between the end period and the period we are valuing at.

SWAPS

Swaps are individual OTC agreements between two companies to exchange their cash flows in a predetermined manner in the future. The two most important types are *interest rate swaps* and *currency swaps*. Swap transactions can also be considered as a portfolio of forward transactions. Their goal is typically to **restructure the balance sheet**.

Interest Rate Swaps

An IRS is a swap where a predetermined fix rate is exchanged for interest at a floating reference rate, both *applied to the same principal amount*. After LIBOR has been phased out, swaps based on overnight rates (SOFR, SONIA) became popular. An Overnight Indexed Swap (OIS) is an agreement to exchange a fixed rate of interest for a reference rate of interest that is calculated from realized overnight rates.

An IRS when first entered into has a *value of zero*. After it has been in existence for some time, its value may be +/- . The notional principal is **not exchanged** in an IRS (the notional is the same amount for both parties).

IRS can be used to transform:

- a liability: a floating-rate loan into a fixed-rate one, or inversely (if company X borrowed at floating-rate + 0.1%, it can enter into a swap where it gets floating-rate and pays fix 3%, netted out it is a fix payment of 3.1%)
- an asset: an asset earning fixed-rate into an asset earning floating-rate, or inversely.

IRSs also entail **comparative advantage**, which means that some companies can borrow at a lower cost at home than companies from abroad, and by a swap they can provide this cheaper borrowing to the other company. Or some companies may have comp. adv. when borrowing in fixed-rate markets, while others in floating rate markets.

Valuation

1) Bond method

The *difference between a variable and a fixed interest rate bond*. Denoting the value of the swap as V , for the party (long) paying fixed interest payments, the value of the swap is:

$$V = B(\text{Floating}) - B(\text{Fixed})$$

Then the procedure is the following:

- 1) Calculate the price of the fixed coupon-bond
- 2) Calculate the price of the floating coupon-bond
- 3) Calculate the difference

The price of the variable-rate bond in the swap transaction corresponds to the face value after each interest payment (thus 100%). Consequently, the fixed-rate bond also reaches its face value, meaning it is issued at par, so the initial value of the fixed bond is also 100%. Bonds that reach face value are called par yield bonds, meaning when IRS is created the fixed rate should be the par yield.

$$\text{Par Yield} = \frac{[1 - DF_T]}{\sum DF_t} * K$$

where K = payments happen in a year.

Cash flows are usually discounted with the LIBOR rate, assuming that the risk of cash flows in the swap transaction is equivalent to the risk of loans provided in the interbank market.

The value of the fixed bond is typically calculated as the present value of cash flows. Between interest payments, the present value of the variable-rate bond equals the present value of the face value and the sum of the present values of variable interest payments until the next payout.

2) FRA method

Each exchange of payment in a swap can be regarded as an FRA, so we can value it by similarly assuming that forward rates are realized. Thus the swap will be a portfolio of FRAs. Then the procedure is the following:

- 1) Calculate floating forward rates
- 2) Calculate the swap cash flows on the assumption that unknown floating rates will equal forward rates
- 3) Discount the swap cash flows at the risk-free rate

Currency Swaps

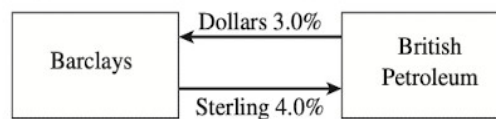
This involves *exchanging principal and interest payments* at a fixed rate in one currency for principal and interest payments at a fixed rate in another currency. A currency swap requires the principal to be specified in each of the two currencies. *The principal amounts in each currency are usually exchanged at the beginning and at the end of the life of the swap.*

CCS can be used to transform:

- a liability: borrowing in one currency into another currency.

Example:

Figure 7.10 A currency swap.

**Pricing of Currency Swaps****1) Bond method:**

The fixed-to-fixed CCS can be valued as the difference of two bonds. The value of the foreign exchange swap for the party paying dollar payment (domestic) and receiving foreign currency is calculated as follows:

$$V = S_0 * B_{\text{€}} - B_{\$}$$

where:

- S is the spot exchange rate,
- $B_{\text{€}}$ is the present value of the euro-denominated fixed bond,
- $B_{\$}$ is the present value of the dollar-denominated fixed bond.

The value of foreign exchange swap transactions is also *zero at inception*. The present values of the bonds are calculated separately in the usual manner.

2) FRA method:

- Determine the forward exchange rates and then break down the swap transaction into a series of forward transactions.
- The value of the swap transaction can always be calculated based on the domestic yield curve and the forward exchange rates.

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- Determine the forward exchange rates. The value of the Forward Rate Agreement (FRA) in dollars for the party paying dollars is:

$$\text{FRA}(i) = \text{PV}(\text{interest rate } \text{€} * F(i) - \text{interest rate } \$).$$

The value of the swap transaction is also:

$$V = \sum \text{FRA}, \text{ which is zero at inception.}$$

The value of a long forward position is always the present value of the difference by which the forward rate exceeds the strike rate. If the interest rates of the two currencies differ and the exchange rate does not change, then for the party paying the lower interest rate, the value of the FRAs for previous periods will be positive, and the value of the FRA representing the final notional exchange will be negative. It's the opposite for the party paying the higher interest rate. Therefore, for the party paying the lower interest rate, the value of the swap transaction will be negative for most of the term.